



**29.9-50K**

# SOLARATOR SERIES

Experience Uninterrupted Power with Solis Energy Storage Inverters

**S6-EH3P(29.9-50)K-H**

Three Phase | High Voltage

## 9 Unique Advantages

- ★ Supports up to 2x rated PV input, maximizing solar energy utilization
- ★ Supports a maximum string input current of 20A, ensuring compatibility with high-power PV modules
- ★ Compatible with 100-280Ah battery modules, reducing overall system costs
- ★ Supports fast battery charging with a maximum charging current of 140A
- ★ Two independent battery ports for flexible configurations and easy capacity expansion
- ★ Delivers 160% overload for 2s in off-grid mode, ensuring stable startup of heavy loads
- ★ Supports generator input across a wide range (20%-100% of rated power), reducing investment costs
- ★ SolisCloud: Smart remote control, AI optimisation, and instant troubleshooting - all in one platform
- ★ 7-inch industrial-grade screen, providing a larger, user-friendly interface for local operation

## 6 Leading Advantages

- Supports both DC and AC coupling, for flexible retrofits and system expansions
- Ensures reliable backup power across diverse scenarios through battery reserve management
- Extends supply time for critical loads with intelligent load prioritization
- Offers a versatile three-in-one interface for seamless integration of on-grid PV, wind power, and diesel generators
- Achieves on- and off-grid transitions in less than 10ms, ensuring an uninterrupted power supply
- Supports multi-unit parallel operation up to 500kW (Solis STS cabinet recommended for systems over 6 units)

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## DATASHEET

| Models                                         | 29.9K              |  | 30K             |  | 40K                                                                                                                                                                                                                                                                                         |  | 50K             |  |
|------------------------------------------------|--------------------|--|-----------------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------------|--|
| Input DC (PV side)                             |                    |  |                 |  |                                                                                                                                                                                                                                                                                             |  |                 |  |
| Recommended max. PV array size                 | 59.8 kW            |  | 60 kW           |  | 80 kW                                                                                                                                                                                                                                                                                       |  | 100 kW          |  |
| Max. usable PV input power                     | 59.8 kW            |  | 60 kW           |  | 80 kW                                                                                                                                                                                                                                                                                       |  | 96 kW           |  |
| Max. input voltage                             |                    |  |                 |  | 1000 V                                                                                                                                                                                                                                                                                      |  |                 |  |
| Rated voltage                                  |                    |  |                 |  | 600 V                                                                                                                                                                                                                                                                                       |  |                 |  |
| Start-up voltage                               |                    |  |                 |  | 180 V                                                                                                                                                                                                                                                                                       |  |                 |  |
| MPPT voltage range                             |                    |  |                 |  | 150 - 850 V                                                                                                                                                                                                                                                                                 |  |                 |  |
| Max. input current                             | 40 A / 40 A / 40 A |  |                 |  |                                                                                                                                                                                                                                                                                             |  | 4 × 40 A        |  |
| Max. short circuit current                     | 60 A / 60 A / 60 A |  |                 |  |                                                                                                                                                                                                                                                                                             |  | 4 × 60 A        |  |
| MPPT number / Max. input strings number        | 3 / 6              |  |                 |  |                                                                                                                                                                                                                                                                                             |  | 4 / 8           |  |
| Battery                                        |                    |  |                 |  |                                                                                                                                                                                                                                                                                             |  |                 |  |
| Battery type                                   |                    |  |                 |  | Li-ion                                                                                                                                                                                                                                                                                      |  |                 |  |
| Battery voltage range                          |                    |  |                 |  | 150 - 800 V                                                                                                                                                                                                                                                                                 |  |                 |  |
| Max. charge / discharge power                  | 32.1 kW            |  | 33 kW           |  | 44 kW                                                                                                                                                                                                                                                                                       |  | 55 kW           |  |
| Max. charge / discharge current                |                    |  |                 |  | 70 A × 2 ①                                                                                                                                                                                                                                                                                  |  |                 |  |
| Number of battery ports                        |                    |  |                 |  | 2                                                                                                                                                                                                                                                                                           |  |                 |  |
| Max. charge / discharge power of each input    | 32.1 kW            |  | 33 kW           |  | 35 kW                                                                                                                                                                                                                                                                                       |  | 35 kW           |  |
| Communication                                  |                    |  |                 |  | CAN / RS485                                                                                                                                                                                                                                                                                 |  |                 |  |
| Output AC (Grid side)                          |                    |  |                 |  |                                                                                                                                                                                                                                                                                             |  |                 |  |
| Rated output power                             | 29.9 kW            |  | 30 kW           |  | 40 kW                                                                                                                                                                                                                                                                                       |  | 50 kW           |  |
| Max. apparent output power                     | 29.9 kVA           |  | 30 kVA          |  | 40 kVA                                                                                                                                                                                                                                                                                      |  | 50 kVA          |  |
| Rated grid voltage                             |                    |  |                 |  | 3/N/PE, 220 V / 380 V, 230 V / 400 V                                                                                                                                                                                                                                                        |  |                 |  |
| Rated grid frequency                           |                    |  |                 |  | 50 Hz / 60 Hz                                                                                                                                                                                                                                                                               |  |                 |  |
| Rated grid output current                      | 45.4 A / 43.2 A    |  | 45.6 A / 43.3 A |  | 60.8 A / 57.7 A                                                                                                                                                                                                                                                                             |  | 76 A / 72.2 A   |  |
| Max. output current                            | 45.4 A / 43.2 A    |  | 45.6 A / 43.3 A |  | 60.8 A / 57.7 A                                                                                                                                                                                                                                                                             |  | 76 A / 72.2 A   |  |
| Power factor                                   |                    |  |                 |  | > 0.99 (0.8 leading - 0.8 lagging)                                                                                                                                                                                                                                                          |  |                 |  |
| THDi                                           |                    |  |                 |  | < 3%                                                                                                                                                                                                                                                                                        |  |                 |  |
| Input AC (Grid side)                           |                    |  |                 |  |                                                                                                                                                                                                                                                                                             |  |                 |  |
| Max. AC passthrough current                    | 90.8 A / 86.4 A    |  | 91.2 A / 86.6 A |  | 121.6 A / 115.4 A                                                                                                                                                                                                                                                                           |  | 152 A / 144.4 A |  |
| Rated input voltage                            |                    |  |                 |  | 3/N/PE, 220 V / 380 V, 230 V / 400 V                                                                                                                                                                                                                                                        |  |                 |  |
| Rated input frequency                          |                    |  |                 |  | 50 Hz / 60 Hz                                                                                                                                                                                                                                                                               |  |                 |  |
| Input Generator                                |                    |  |                 |  |                                                                                                                                                                                                                                                                                             |  |                 |  |
| Max. input power                               | 29.9 kW            |  | 30 kW           |  | 40 kW                                                                                                                                                                                                                                                                                       |  | 50 kW           |  |
| Rated input current                            | 45.4 A / 43.2 A    |  | 45.6 A / 43.3 A |  | 60.8 A / 57.7 A                                                                                                                                                                                                                                                                             |  | 76 A / 72.2 A   |  |
| Rated input voltage                            |                    |  |                 |  | 3/N/PE, 220 V / 380 V, 230 V / 400 V                                                                                                                                                                                                                                                        |  |                 |  |
| Rated input frequency                          |                    |  |                 |  | 50 Hz / 60 Hz                                                                                                                                                                                                                                                                               |  |                 |  |
| Output AC (Back-up)                            |                    |  |                 |  |                                                                                                                                                                                                                                                                                             |  |                 |  |
| Rated output power                             | 29.9 kW            |  | 30 kW           |  | 40 kW                                                                                                                                                                                                                                                                                       |  | 50 kW           |  |
| Max. apparent output power                     |                    |  |                 |  | 1.6 times of rated power, 2 s                                                                                                                                                                                                                                                               |  |                 |  |
| Back-up switch time                            |                    |  |                 |  | < 10 ms                                                                                                                                                                                                                                                                                     |  |                 |  |
| Rated output voltage                           |                    |  |                 |  | 3/N/PE, 220 V / 380 V, 230 V / 400 V                                                                                                                                                                                                                                                        |  |                 |  |
| Rated frequency                                |                    |  |                 |  | 50 Hz / 60 Hz                                                                                                                                                                                                                                                                               |  |                 |  |
| Rated output current                           | 45.4 A / 43.2 A    |  | 45.6 A / 43.3 A |  | 60.8 A / 57.7 A                                                                                                                                                                                                                                                                             |  | 76 A / 72.2 A   |  |
| THDv (@linear load)                            |                    |  |                 |  | < 2%                                                                                                                                                                                                                                                                                        |  |                 |  |
| Efficiency                                     |                    |  |                 |  |                                                                                                                                                                                                                                                                                             |  |                 |  |
| Max. efficiency                                |                    |  |                 |  | 97.8%                                                                                                                                                                                                                                                                                       |  |                 |  |
| EU efficiency                                  |                    |  |                 |  | 97.4%                                                                                                                                                                                                                                                                                       |  |                 |  |
| BAT charged by PV max. efficiency              |                    |  |                 |  | 98.5%                                                                                                                                                                                                                                                                                       |  |                 |  |
| BAT charged / discharged to AC max. efficiency |                    |  |                 |  | 97.5%                                                                                                                                                                                                                                                                                       |  |                 |  |
| Protection                                     |                    |  |                 |  |                                                                                                                                                                                                                                                                                             |  |                 |  |
| Anti-islanding protection                      |                    |  |                 |  | Yes                                                                                                                                                                                                                                                                                         |  |                 |  |
| Output over current protection                 |                    |  |                 |  | Yes                                                                                                                                                                                                                                                                                         |  |                 |  |
| Short circuit protection                       |                    |  |                 |  | Yes                                                                                                                                                                                                                                                                                         |  |                 |  |
| Integrated DC switch                           |                    |  |                 |  | Yes                                                                                                                                                                                                                                                                                         |  |                 |  |
| DC reverse-polarity protection                 |                    |  |                 |  | Yes                                                                                                                                                                                                                                                                                         |  |                 |  |
| Surge protection                               |                    |  |                 |  | DC Type II / AC Type II                                                                                                                                                                                                                                                                     |  |                 |  |
| Integrated AFCI 2.0                            |                    |  |                 |  | Optional                                                                                                                                                                                                                                                                                    |  |                 |  |
| General Data                                   |                    |  |                 |  |                                                                                                                                                                                                                                                                                             |  |                 |  |
| Dimensions (W × H × D)                         |                    |  |                 |  | 530 × 880 × 290 mm                                                                                                                                                                                                                                                                          |  |                 |  |
| Weight                                         |                    |  |                 |  | 73 kg                                                                                                                                                                                                                                                                                       |  |                 |  |
| Topology                                       |                    |  |                 |  | Transformerless                                                                                                                                                                                                                                                                             |  |                 |  |
| Self-consumption (night)                       |                    |  |                 |  | < 35 W                                                                                                                                                                                                                                                                                      |  |                 |  |
| Operating ambient temperature range            |                    |  |                 |  | -25 ~ +60°C                                                                                                                                                                                                                                                                                 |  |                 |  |
| Relative humidity                              |                    |  |                 |  | 0 - 100%                                                                                                                                                                                                                                                                                    |  |                 |  |
| Ingress protection                             |                    |  |                 |  | IP66                                                                                                                                                                                                                                                                                        |  |                 |  |
| Cooling concept                                |                    |  |                 |  | Intelligent fan-cooling                                                                                                                                                                                                                                                                     |  |                 |  |
| Max. operation altitude                        |                    |  |                 |  | 4000 m                                                                                                                                                                                                                                                                                      |  |                 |  |
| Grid connection standard                       |                    |  |                 |  | G99, VDE-AR-N 4105/VDE V 0124, EN 50549-1/EN 50549-10, VDE 0126/UTE C 15/VFR:2019, NTS 631/RD 1699/RD 244/UNE 206006/UNE 206007-1, CEI 0-21, C10/11, NRS 097-2-1, TOR, EIFS 2018.2, IEC 62116, IEC 61727, IEC 60068, IEC 61683, EN 50530, MEA, PEA, PORTARIA Nº 140, DE 21 DE MARÇO DE 2022 |  |                 |  |
| Safety / EMC standard                          |                    |  |                 |  | IEC/EN 62109-1/-2, IEC/EN 61000-6-2/-4, EN 55011                                                                                                                                                                                                                                            |  |                 |  |
| Features                                       |                    |  |                 |  |                                                                                                                                                                                                                                                                                             |  |                 |  |
| PV connection                                  |                    |  |                 |  | MC4 Quick connection plug                                                                                                                                                                                                                                                                   |  |                 |  |
| Battery connection                             |                    |  |                 |  | Terminal connector                                                                                                                                                                                                                                                                          |  |                 |  |
| AC connection                                  |                    |  |                 |  | Terminal Block                                                                                                                                                                                                                                                                              |  |                 |  |
| Display                                        |                    |  |                 |  | 7.0" LCD display & Bluetooth + APP                                                                                                                                                                                                                                                          |  |                 |  |
| Communication                                  |                    |  |                 |  | CAN, RS485, Ethernet, Optional: Wi-Fi, Cellular, LAN                                                                                                                                                                                                                                        |  |                 |  |

① Supporting parallel 140A input.